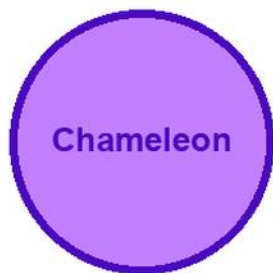


KITABU QUEST

S6

CHEMISTRY



Cover scene

students doing safe chemistry work with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

6

Title Page

KITABU QUEST S6 Chemistry

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S6 learners study Chemistry one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use safe observations, simple models, data tables, and evidence-based explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Properties and uses of transition metals
2. Extraction of metals
3. NPK as components of fertilizers
4. Benzene
5. Derivatives of benzene
6. Polymers and polymerisation
7. Solvent extraction and colligative properties
8. Quantitative chemical equilibrium
9. pH of acidic and alkaline solutions
10. Indicators and titration curves
11. Solubility and solubility product for sparingly soluble salts
12. Electrochemical cell and applications

As you finish each topic, return to the success criteria and correct one answer before moving on.

Properties and uses of transition metals: Lesson Opener

Learning focus: Properties and uses of transition metals

Country link: a safe Rwanda observation, model, data table, or investigation for Properties and uses of transition metals.

By the end of this topic, you should be able to:

- the explain the properties and uses of transition metals
- discuss qualitatively the properties of transition elements when compared to those of calcium as a typical s-block metal
- explain the principle of ligand exchange
- state the rules of naming complex ions and stereo isomerism
- describe reactions of transition metals
- state the uses of transition metals

Inquiry questions:

- How can properties and uses of transition metals help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of transition metals: Learn

Properties and uses of transition metals is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Properties and uses of transition metals
- Properties
- uses
- transition
- metals
- concept
- observation
- evidence

Properties and uses of transition metals: Worked Example

Investigation model for Properties and uses of transition metals: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Properties and uses of transition metals: Vocabulary And Study Skill

Use these words and study moves before you practise Properties and uses of transition metals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Properties and uses of transition metals
- Properties
- uses
- transition
- metals
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Properties and uses of transition metals without a clear example, method, or evidence.

Properties and uses of transition metals: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Properties and uses of transition metals.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Properties and uses of transition metals: Practice And Correction

Practice for Properties and uses of transition metals:

Main outcome: the explain the properties and uses of transition metals.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Properties and uses of transition metals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Properties and uses of transition metals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of transition metals: Outcome Check

Outcome check for Properties and uses of transition metals: The explain the properties and uses of transition metals.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of transition metals: Outcome Check

Outcome check for Properties and uses of transition metals: Discuss qualitatively the properties of transition elements when compared to those of calcium as a typical s-block metal.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Extraction of metals: Lesson Opener

Learning focus: Extraction of metals

Country link: a safe Rwanda observation, model, data table, or investigation for Extraction of metals.

By the end of this topic, you should be able to:

- the relate the properties of metals to their methods of extraction and uses, and suggest preventive measures to the dangers associated with their extraction
- describe the extraction of copper, iron, sodium, tantalum, zinc, wolfram, aluminium and tantalum
- outline and explain the uses copper, iron, tantalum, zinc, wolfram, tin ore (cassiterite)
- explain the issues associated with the extraction of metals and preventive measures
- relate the properties of metals to their methods of extraction
- appreciate the use of transitional metals

Inquiry questions:

- How can extraction of metals help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Extraction of metals: Learn

Extraction of metals is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Extraction of metals
- Extraction
- metals
- concept
- observation
- evidence
- safety
- conclusion

Extraction of metals: Worked Example

Investigation model for Extraction of metals: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Extraction of metals: Vocabulary And Study Skill

Use these words and study moves before you practise Extraction of metals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Extraction of metals
- Extraction
- metals
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Extraction of metals without a clear example, method, or evidence.

Extraction of metals: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Extraction of metals.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Extraction of metals: Practice And Correction

Practice for Extraction of metals:

Main outcome: the relate the properties of metals to their methods of extraction and uses, and suggest preventive measures to the dangers associated with their extraction.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Extraction of metals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Extraction of metals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Extraction of metals: Outcome Check

Outcome check for Extraction of metals: The relate the properties of metals to their methods of extraction and uses, and suggest preventive measures to the dangers associated with their extraction.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Extraction of metals: Outcome Check

Outcome check for Extraction of metals: Describe the extraction of copper, iron, sodium, tantalum, zinc, wolfram, aluminium and tantalum.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

NPK as components of fertilizers: Lesson Opener

Learning focus: NPK as components of fertilizers

Country link: a safe Rwanda observation, model, data table, or investigation for NPK as components of fertilizers.

By the end of this topic, you should be able to:

- the analyse the components of quality fertilizers and their benefits, effects of misuse and dangers associated with the substandard fertilizers
- state the major constituents of the fertilizers
- identify the characteristics of a good fertilizer
- state the chemical reactions involved in the manufacture of the fertilizers
- briefly describe the manufacture of the fertilizers
- state the advantages or disadvantages of using the fertilizers

Inquiry questions:

- How can npk as components of fertilizers help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

NPK as components of fertilizers: Learn

NPK as components of fertilizers is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- NPK as components of fertilizers
- components
- fertilizers
- concept
- observation
- evidence
- safety
- conclusion

NPK as components of fertilizers: Worked Example

Investigation model for NPK as components of fertilizers: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

NPK as components of fertilizers: Vocabulary And Study Skill

Use these words and study moves before you practise NPK as components of fertilizers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- NPK as components of fertilizers
- components
- fertilizers
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about NPK as components of fertilizers without a clear example, method, or evidence.

NPK as components of fertilizers: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for NPK as components of fertilizers.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

NPK as components of fertilizers: Practice And Correction

Practice for NPK as components of fertilizers:

Main outcome: the analyse the components of quality fertilizers and their benefits, effects of misuse and dangers associated with the substandard fertilizers.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

NPK as components of fertilizers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from NPK as components of fertilizers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

NPK as components of fertilizers: Outcome Check

Outcome check for NPK as components of fertilizers: The analyse the components of quality fertilizers and their benefits, effects of misuse and dangers associated with the substandard fertilizers.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

NPK as components of fertilizers: Outcome Check

Outcome check for NPK as components of fertilizers: State the major constituents of the fertilizers.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Benzene: Lesson Opener

Learning focus: Benzene

Country link: a safe Rwanda observation, model, data table, or investigation for Benzene.

By the end of this topic, you should be able to:

- the relate the chemistry and uses of benzene to its nature and structure
- state the physical properties of benzene
- describe the uses of benzene
- outline the preparations of benzene
- relate the conditions for reactions of benzene to its chemical stability

Inquiry questions:

- How can benzene help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Benzene: Learn

Benzene introduces benzene as an aromatic hydrocarbon with a special ring structure.

Core idea:

- Benzene has six carbon atoms and six hydrogen atoms in a ring.
- Its bonding is represented by alternating double bonds or a circle inside the ring to show delocalised electrons.
- Benzene reactions are often substitution reactions rather than simple addition reactions.
- Benzene and many organic solvents must be handled only through teacher-approved demonstrations because safety matters.
- Structure, formula, properties, and safe-use rules should be connected, not memorised separately.

Key words:

- Benzene
- concept
- observation
- evidence
- safety
- conclusion

Benzene: Worked Example

Worked example for Benzene: Interpret a benzene ring diagram.

1. Count six carbon positions in the ring.
2. Add one hydrogen attached to each carbon to give C_6H_6 .
3. Explain that the ring can be drawn with alternating double bonds or a circle showing delocalised electrons.
4. State one reaction idea: benzene often undergoes substitution reactions.
5. Add one safety rule for aromatic compounds and solvents.

Conclusion: benzene's ring structure explains why its reactions differ from many alkenes.

Benzene: Vocabulary And Study Skill

Use these words and study moves before you practise Benzene. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Benzene
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Benzene without a clear example, method, or evidence.

Benzene: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Benzene.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Benzene: Practice And Correction

Practice for Benzene:

Main outcome: the relate the chemistry and uses of benzene to its nature and structure.

1. Draw or interpret a benzene ring representation.
2. State the formula C_6H_6 and one feature of delocalised electrons.
3. Explain one safe handling rule for organic compounds or solvents.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Benzene: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Benzene.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Benzene: Outcome Check

Outcome check for Benzene: The relate the chemistry and uses of benzene to its nature and structure.

1. Draw or identify the benzene ring.
2. State the formula C_6H_6 .
3. Link ring structure to one reaction or safety point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Benzene: Outcome Check

Outcome check for Benzene: State the physical properties of benzene.

1. Draw or identify the benzene ring.
2. State the formula C_6H_6 .
3. Link ring structure to one reaction or safety point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Derivatives of benzene: Lesson Opener

Learning focus: Derivatives of benzene

Country link: a safe Rwanda observation, model, data table, or investigation for Derivatives of benzene.

By the end of this topic, you should be able to:

- the relate aromatic ketones, aldehydes, carboxylic acids and amines to their chemical activity
- explain the effects of substituent groups on the benzene ring
- give systematic names of aromatic compounds
- describe the preparation and reactions of phenol, benzoic acid, benzaldehyde, phenyl ethanone and phenylamine
- state the uses of phenols
- describe the reaction of phenol, aromatic carbonyl compounds and carboxylic acids

Inquiry questions:

- How can derivatives of benzene help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Derivatives of benzene: Learn

Derivatives of benzene introduces benzene as an aromatic hydrocarbon with a special ring structure.

Core idea:

- Benzene has six carbon atoms and six hydrogen atoms in a ring.
- Its bonding is represented by alternating double bonds or a circle inside the ring to show delocalised electrons.
- Benzene reactions are often substitution reactions rather than simple addition reactions.
- Benzene and many organic solvents must be handled only through teacher-approved demonstrations because safety matters.
- Structure, formula, properties, and safe-use rules should be connected, not memorised separately.

Key words:

- Derivatives of benzene
- Derivatives
- benzene
- concept
- observation
- evidence
- safety
- conclusion

Derivatives of benzene: Worked Example

Worked example for Derivatives of benzene: Interpret a benzene ring diagram.

1. Count six carbon positions in the ring.
2. Add one hydrogen attached to each carbon to give C₆H₆.
3. Explain that the ring can be drawn with alternating double bonds or a circle showing delocalised electrons.
4. State one reaction idea: benzene often undergoes substitution reactions.
5. Add one safety rule for aromatic compounds and solvents.

Conclusion: benzene's ring structure explains why its reactions differ from many alkenes.

Derivatives of benzene: Vocabulary And Study Skill

Use these words and study moves before you practise Derivatives of benzene. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Derivatives of benzene
- Derivatives
- benzene
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Derivatives of benzene without a clear example, method, or evidence.

Derivatives of benzene: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Derivatives of benzene.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Derivatives of benzene: Practice And Correction

Practice for Derivatives of benzene:

Main outcome: the relate aromatic ketones, aldehydes, carboxylic acids and amines to their chemical activity.

1. Draw or interpret a benzene ring representation.
2. State the formula C_6H_6 and one feature of delocalised electrons.
3. Explain one safe handling rule for organic compounds or solvents.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Derivatives of benzene: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Derivatives of benzene.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Derivatives of benzene: Outcome Check

Outcome check for Derivatives of benzene: The relate aromatic ketones, aldehydes, carboxylic acids and amines to their chemical activity.

1. Draw or identify the benzene ring.
2. State the formula C_6H_6 .
3. Link ring structure to one reaction or safety point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Derivatives of benzene: Outcome Check

Outcome check for Derivatives of benzene: Explain the effects of substituent groups on the benzene ring.

1. Draw or identify the benzene ring.
2. State the formula C_6H_6 .
3. Link ring structure to one reaction or safety point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Polymers and polymerisation: Lesson Opener

Learning focus: Polymers and polymerisation

Country link: a safe Rwanda observation, model, data table, or investigation for Polymers and polymerisation.

By the end of this topic, you should be able to:

- the relate the types of polymers to their structural properties and uses
- define the terms monomer, polymer and polymerisation
- describe the formation of polymers
- describe addition and condensation of polymerisation
- explain the terms thermosetting and thermo-softening of the plastics
- discuss the advantages and disadvantages of both natural and synthetic polymers

Inquiry questions:

- How can polymers and polymerisation help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Polymers and polymerisation: Learn

Polymers and polymerisation is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Polymers and polymerisation
- Polymers
- polymerisation
- concept
- observation
- evidence
- safety
- conclusion

Polymers and polymerisation: Worked Example

Investigation model for Polymers and polymerisation: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Polymers and polymerisation: Vocabulary And Study Skill

Use these words and study moves before you practise Polymers and polymerisation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Polymers and polymerisation
- Polymers
- polymerisation
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Polymers and polymerisation without a clear example, method, or evidence.

Polymers and polymerisation: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Polymers and polymerisation.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Polymers and polymerisation: Practice And Correction

Practice for Polymers and polymerisation:

Main outcome: the relate the types of polymers to their structural properties and uses.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Polymers and polymerisation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Polymers and polymerisation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Polymers and polymerisation: Outcome Check

Outcome check for Polymers and polymerisation: The relate the types of polymers to their structural properties and uses.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Polymers and polymerisation: Outcome Check

Outcome check for Polymers and polymerisation: Define the terms monomer, polymer and polymerisation.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solvent extraction and colligative properties: Lesson Opener

Learning focus: Solvent extraction and colligative properties

Country link: a safe Rwanda observation, model, data table, or investigation for Solvent extraction and colligative properties.

By the end of this topic, you should be able to:

- the learner should be able to apply partition and Raoult's law to separate mixtures and determine the molecular and formula masses of compounds using colligative properties
- define partition coefficient (distribution law) and solvent extraction
- state the Raoult's law
- state and explain the advantages of carrying out distillation processes under reduced pressures
- discuss the chemical principles upon which simple distillation, fractional distillation and steam distillation are
- carry out simple separation experiments based on the solute partitioning between two immiscible solvents

Inquiry questions:

- How can solvent extraction and colligative properties help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solvent extraction and colligative properties: Learn

Solvent extraction and colligative properties teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Solvent extraction and colligative properties
- Solvent
- extraction
- colligative
- properties
- concept
- observation
- evidence

Solvent extraction and colligative properties: Worked Example

Worked example for Solvent extraction and colligative properties: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Solvent extraction and colligative properties: Vocabulary And Study Skill

Use these words and study moves before you practise Solvent extraction and colligative properties. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solvent extraction and colligative properties
- Solvent
- extraction
- colligative
- properties
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solvent extraction and colligative properties without a clear example, method, or evidence.

Solvent extraction and colligative properties: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Solvent extraction and colligative properties.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Solvent extraction and colligative properties: Practice And Correction

Practice for Solvent extraction and colligative properties:

Main outcome: the learner should be able to apply partition and Raoult's law to separate mixtures and determine the molecular and formula masses of compounds using colligative properties.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Solvent extraction and colligative properties: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solvent extraction and colligative properties.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solvent extraction and colligative properties: Outcome Check

Outcome check for Solvent extraction and colligative properties: The learner should be able to apply partition and Raoult's law to separate mixtures and determine the molecular and formula masses of compounds using colligative properties.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solvent extraction and colligative properties: Outcome Check

Outcome check for Solvent extraction and colligative properties: Define partition coefficient (distribution law) and solvent extraction.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative chemical equilibrium: Lesson Opener

Learning focus: Quantitative chemical equilibrium

Country link: a safe Rwanda observation, model, data table, or investigation for Quantitative chemical equilibrium.

By the end of this topic, you should be able to:

- the write expressions and calculate the values of equilibrium constant, interpret the
- explain how the temperature affects the magnitude of equilibrium constant K_c
- derive the relationship between K_c and K_p
- derive equilibrium constant K_c
- interpret the K_c the yield of the reversible reactions
- investigate and explain quantitative chemical equilibrium using evidence

Inquiry questions:

- How can quantitative chemical equilibrium help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative chemical equilibrium: Learn

Quantitative chemical equilibrium is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Quantitative chemical equilibrium
- Quantitative
- chemical
- equilibrium
- concept
- observation
- evidence
- safety

Quantitative chemical equilibrium: Worked Example

Investigation model for Quantitative chemical equilibrium: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Quantitative chemical equilibrium: Vocabulary And Study Skill

Use these words and study moves before you practise Quantitative chemical equilibrium. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Quantitative chemical equilibrium
- Quantitative
- chemical
- equilibrium
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Quantitative chemical equilibrium without a clear example, method, or evidence.

Quantitative chemical equilibrium: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Quantitative chemical equilibrium.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Quantitative chemical equilibrium: Practice And Correction

Practice for Quantitative chemical equilibrium:

Main outcome: the write expressions and calculate the values of equilibrium constant, interpret the.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Quantitative chemical equilibrium: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Quantitative chemical equilibrium.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative chemical equilibrium: Outcome Check

Outcome check for Quantitative chemical equilibrium: The write expressions and calculate the values of equilibrium constant, interpret the.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative chemical equilibrium: Outcome Check

Outcome check for Quantitative chemical equilibrium: Explain how the temperature affects the magnitude of equilibrium constant K_c .

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

pH of acidic and alkaline solutions: Lesson Opener

Learning focus: pH of acidic and alkaline solutions

Country link: a safe Rwanda observation, model, data table, or investigation for pH of acidic and alkaline solutions.

By the end of this topic, you should be able to:

- the prepare solutions, measure their pH and calculate the pH of acidic and alkaline solutions.

Explain the concept of buffer solution, hydrolysis of salts and discuss its applications in manufacturing industry and biological processes

- define the degree of ionization (α)
- define the terms K_a , pH, pK_a , K_b , pK_b and K_w
- write equations for salt hydrolysis reactions and the expression for the hydrolysis constant
- define the term buffer solution
- explain how buffer solutions control pH

Inquiry questions:

- How can pH of acidic and alkaline solutions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

pH of acidic and alkaline solutions: Learn

pH of acidic and alkaline solutions is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- pH of acidic and alkaline solutions
- acidic
- alkaline
- solutions
- concent

pH of acidic and alkaline solutions: Worked Example

Investigation model for pH of acidic and alkaline solutions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

pH of acidic and alkaline solutions: Vocabulary And Study Skill

Use these words and study moves before you practise pH of acidic and alkaline solutions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- pH of acidic and alkaline solutions
- acidic
- alkaline
- solutions
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about pH of acidic and alkaline solutions without a clear example, method, or evidence.

pH of acidic and alkaline solutions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for pH of acidic and alkaline solutions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

pH of acidic and alkaline solutions: Practice And Correction

Practice for pH of acidic and alkaline solutions:

Main outcome: the prepare solutions, measure their pH and calculate the pH of acidic and alkaline solutions. Explain the concept of buffer solution, hydrolysis of salts and discuss its applications in manufacturing industry and biological processes.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

pH of acidic and alkaline solutions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from pH of acidic and alkaline solutions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

pH of acidic and alkaline solutions: Outcome Check

Outcome check for pH of acidic and alkaline solutions: The prepare solutions, measure their pH and calculate the pH of acidic and alkaline solutions. Explain the concept of buffer solution, hydrolysis of salts and discuss its applications in manufacturing industry and biological processes.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

pH of acidic and alkaline solutions: Outcome Check

Outcome check for pH of acidic and alkaline solutions: Define the degree of ionization (α).

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Indicators and titration curves: Lesson Opener

Learning focus: Indicators and titration curves

Country link: a safe Rwanda observation, model, data table, or investigation for Indicators and titration curves.

By the end of this topic, you should be able to:

- the relate titration curves to the type of acid and base titrated, properly choose and use indicators in acid-base titrations
- define the term indicator
- explain how the indicators work
- explain what is meant by the pH range of indicator
- state the criteria for the selection of acid-base indicator for the use in titrations
- describe the changes in pH during acid/base titrations

Inquiry questions:

- How can indicators and titration curves help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Indicators and titration curves: Learn

Indicators and titration curves is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Indicators and titration curves
- Indicators
- titration
- curves
- concept
- observation
- evidence
- safety

Indicators and titration curves: Worked Example

Investigation model for Indicators and titration curves: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Indicators and titration curves: Vocabulary And Study Skill

Use these words and study moves before you practise Indicators and titration curves. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Indicators and titration curves
- Indicators
- titration
- curves
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Indicators and titration curves without a clear example, method, or evidence.

Indicators and titration curves: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Indicators and titration curves.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Indicators and titration curves: Practice And Correction

Practice for Indicators and titration curves:

Main outcome: the relate titration curves to the type of acid and base titrated, properly choose and use indicators in acid-base titrations.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Indicators and titration curves: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Indicators and titration curves.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Indicators and titration curves: Outcome Check

Outcome check for Indicators and titration curves: The relate titration curves to the type of acid and base titrated, properly choose and use indicators in acid-base titrations.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Indicators and titration curves: Outcome Check

Outcome check for Indicators and titration curves: Define the term indicator.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solubility and solubility product for sparingly soluble salts: Lesson Opener

Learning focus: Solubility and solubility product for sparingly soluble salts

Country link: a safe Rwanda observation, model, data table, or investigation for Solubility and solubility product for sparingly soluble salts.

By the end of this topic, you should be able to:

- the calculate the solubility product, constant of sparingly soluble salts, and deduce the applications of common ion effect in the industry
- define the term solubility product K_{sp}
- state and explain the factors that affect solubility of sparingly soluble salts
- state and explain the applications of solubility product
- perform a simple experiment to determine the solubility product of a sparingly soluble salt
- write the equations of dissociation and K_{sp} expression for sparingly soluble salts

Inquiry questions:

- How can solubility and solubility product for sparingly soluble salts help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solubility and solubility product for sparingly soluble salts: Learn

Solubility and solubility product for sparingly soluble salts is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Solubility and solubility product for sparingly soluble salts
- Solubility
- solubility
- product
- sparingly
- soluble
- salts
- concept

Solubility and solubility product for sparingly soluble salts: Worked Example

Investigation model for Solubility and solubility product for sparingly soluble salts: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Solubility and solubility product for sparingly soluble salts: Vocabulary And Study Skill

Use these words and study moves before you practise Solubility and solubility product for sparingly soluble salts. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solubility and solubility product for sparingly soluble salts
- Solubility
- solubility
- product
- sparingly
- soluble
- salts
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solubility and solubility product for sparingly soluble salts without a clear example, method, or evidence.

Solubility and solubility product for sparingly soluble salts: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Solubility and solubility product for sparingly soluble salts.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Solubility and solubility product for sparingly soluble salts: Practice And Correction

Practice for Solubility and solubility product for sparingly soluble salts:

Main outcome: the calculate the solubility product, constant of sparingly soluble salts, and deduce the applications of common ion effect in the industry.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Solubility and solubility product for sparingly soluble salts: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solubility and solubility product for sparingly soluble salts.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solubility and solubility product for sparingly soluble salts: Outcome Check

Outcome check for Solubility and solubility product for sparingly soluble salts: The calculate the solubility product, constant of sparingly soluble salts, and deduce the applications of common ion effect in the industry.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Solubility and solubility product for sparingly soluble salts: Outcome Check

Outcome check for Solubility and solubility product for sparingly soluble salts: Define the term solubility product K_{sp} .

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrochemical cell and applications: Lesson Opener

Learning focus: Electrochemical cell and applications

Country link: a safe Rwanda observation, model, data table, or investigation for Electrochemical cell and applications.

By the end of this topic, you should be able to:

- the explain the working and industrial applications of electrochemical and electrolytic cells
- define the term electrochemical
- describe the standard hydrogen electrode
- explain the working of galvanic cells using the fully labelled diagram
- describe industrial applications of electrochemical cells
- construct a simple galvanic cell

Inquiry questions:

- How can electrochemical cell and applications help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrochemical cell and applications: Learn

Electrochemical cell and applications teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Electrochemical cell and applications
- Electrochemical
- cell
- applications
- concept

Electrochemical cell and applications: Worked Example

Worked example for Electrochemical cell and applications: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Electrochemical cell and applications: Vocabulary And Study Skill

Use these words and study moves before you practise Electrochemical cell and applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electrochemical cell and applications
- Electrochemical
- cell
- applications
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electrochemical cell and applications without a clear example, method, or evidence.

Electrochemical cell and applications: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Electrochemical cell and applications.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electrochemical cell and applications: Practice And Correction

Practice for Electrochemical cell and applications:

Main outcome: the explain the working and industrial applications of electrochemical and electrolytic cells.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electrochemical cell and applications: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electrochemical cell and applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrochemical cell and applications: Outcome Check

Outcome check for Electrochemical cell and applications: The explain the working and industrial applications of electrochemical and electrolytic cells.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrochemical cell and applications: Outcome Check

Outcome check for Electrochemical cell and applications: Define the term electrochemical.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect the rate of reactions: Lesson Opener

Learning focus: Factors that affect the rate of reactions

Country link: a safe Rwanda observation, model, data table, or investigation for Factors that affect the rate of reactions.

By the end of this topic, you should be able to:

- the explain the factors that affect the rate of chemical reaction and use Arrhenius equation to calculate the ratio of rate constant and activation energy with change in the temperature
- explain the concept of reaction kinetics
- explain the effect of different conditions on the rate of reaction
- carry out experiments to show how different factors affect the rate of chemical reactions. Predict the effect of changing conditions on the rate of reactions
- appreciate the importance of reaction kinetics
- appreciate the importance of different conditions on the reaction rates

Inquiry questions:

- How can factors that affect the rate of reactions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect the rate of reactions: Learn

Factors that affect the rate of reactions is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Factors that affect the rate of reactions
- Factors
- that
- affect
- rate
- reactions
- concept
- observation

Factors that affect the rate of reactions: Worked Example

Investigation model for Factors that affect the rate of reactions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Factors that affect the rate of reactions: Vocabulary And Study Skill

Use these words and study moves before you practise Factors that affect the rate of reactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Factors that affect the rate of reactions
- Factors
- that
- affect
- rate
- reactions
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Factors that affect the rate of reactions without a clear example, method, or evidence.

Factors that affect the rate of reactions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Factors that affect the rate of reactions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Factors that affect the rate of reactions: Practice And Correction

Practice for Factors that affect the rate of reactions:

Main outcome: the explain the factors that affect the rate of chemical reaction and use Arrhenius equation to calculate the ratio of rate constant and activation energy with change in the temperature.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Factors that affect the rate of reactions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Factors that affect the rate of reactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect the rate of reactions: Outcome Check

Outcome check for Factors that affect the rate of reactions: The explain the factors that affect the rate of chemical reaction and use Arrhenius equation to calculate the ratio of rate constant and activation energy with change in the temperature.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect the rate of reactions: Outcome Check

Outcome check for Factors that affect the rate of reactions: Explain the concept of reaction kinetics.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rate laws and measurements: Lesson Opener

Learning focus: Rate laws and measurements

Country link: a safe Rwanda observation, model, data table, or investigation for Rate laws and measurements.

By the end of this topic, you should be able to:

- the measure the rates of reaction and formulate simple rate equations using the experimental results
- state and explain kinetic conditions for a chemical reaction to take place
- explain the effect of the temperature and catalysts on the rate of the reaction using Boltzmann distribution of energies (and of collision frequency)
- differentiate between SN1 and SN2 mechanisms
- state and explain the rate determining the steps for multi-step reactions
- deduce the order of reaction from appropriate experimental data

Inquiry questions:

- How can rate laws and measurements help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rate laws and measurements: Learn

Rate laws and measurements begins with safe behaviour and accurate measurement.

Core idea:

- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Rwanda, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:

- Rate laws and measurements
- Rate
- laws
- measurements
- concept
- observation
- evidence
- safety

Rate laws and measurements: Worked Example

Investigation model for Rate laws and measurements: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Rate laws and measurements: Vocabulary And Study Skill

Use these words and study moves before you practise Rate laws and measurements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Rate laws and measurements
- Rate
- laws
- measurements
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Rate laws and measurements without a clear example, method, or evidence.

Rate laws and measurements: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Rate laws and measurements.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Rate laws and measurements: Practice And Correction

Practice for Rate laws and measurements:

Main outcome: the measure the rates of reaction and formulate simple rate equations using the experimental results.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Rate laws and measurements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Rate laws and measurements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rate laws and measurements: Outcome Check

Outcome check for Rate laws and measurements: The measure the rates of reaction and formulate simple rate equations using the experimental results.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Rate laws and measurements: Outcome Check

Outcome check for Rate laws and measurements: State and explain kinetic conditions for a chemical reaction to take place.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity: Lesson Opener

Learning focus: Radioactivity

Country link: a safe Rwanda observation, model, data table, or investigation for Radioactivity.

By the end of this topic, you should be able to:

- the explain the importance and dangers of radioisotopes in everyday life
- explain the process of radioactivity
- explain the properties of alpha, beta and gamma rays
- explain half-lives of radioactive radioisotopes
- explain the applications of radioisotopes in medicine, agriculture and industries
- compare and contrast chemical and nuclear reactions

Inquiry questions:

- How can radioactivity help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity: Learn

Radioactivity explains radioactivity as the process where unstable nuclei emit radiation to become more stable.

Core idea:

- Alpha, beta, and gamma radiation differ in charge, mass, penetration, and shielding.
- Half-life is the time taken for half of a radioactive sample or count rate to remain.
- Radioactivity can be useful in medicine, industry, agriculture, dating, and research when controlled safely.
- Radiation can also damage living tissue, so distance, shielding, time limits, and trained supervision are essential.
- Learners should interpret simple count-rate data and avoid unsafe experiments.

Key words:

- Radioactivity
- concept
- observation
- evidence
- safety
- conclusion

Radioactivity: Worked Example

Worked example for Radioactivity: Use half-life data.

A sample has a count rate of 80 counts per minute. Its half-life is 5 minutes.

1. After 5 minutes, half remains: 40 counts per minute.
2. After 10 minutes, half of 40 remains: 20 counts per minute.
3. After 15 minutes, half of 20 remains: 10 counts per minute.
4. Shielding and distance must be controlled by trained adults.

Answer: after 15 minutes, the count rate is 10 counts per minute.

Radioactivity: Vocabulary And Study Skill

Use these words and study moves before you practise Radioactivity. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Radioactivity
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Radioactivity without a clear example, method, or evidence.

Radioactivity: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Radioactivity.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Radioactivity: Practice And Correction

Practice for Radioactivity:

Main outcome: the explain the importance and dangers of radioisotopes in everyday life.

1. Compare alpha, beta, and gamma radiation in a table.
2. Solve one half-life count-rate question.
3. Write two radiation safety rules involving time, distance, or shielding.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Radioactivity: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Radioactivity.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity: Outcome Check

Outcome check for Radioactivity: The explain the importance and dangers of radioisotopes in everyday life.

1. Distinguish alpha, beta, and gamma radiation.
2. Use half-life in one simple calculation.
3. Write one radiation safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity: Outcome Check

Outcome check for Radioactivity: Explain the process of radioactivity.

1. Distinguish alpha, beta, and gamma radiation.
2. Use half-life in one simple calculation.
3. Write one radiation safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of transition metals: Review Clinic 1

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Properties and uses of transition metals that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Extraction of metals: Review Clinic 2

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Extraction of metals that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

NPK as components of fertilizers: Review Clinic 3

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from NPK as components of fertilizers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Benzene: Review Clinic 4

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Benzene that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Chemistry.

- Properties and uses of transition metals: Explain this term using an example from the book, your school, or your community.
- Extraction of metals: Explain this term using an example from the book, your school, or your community.
- NPK as components of fertilizers: Explain this term using an example from the book, your school, or your community.
- Benzene: Explain this term using an example from the book, your school, or your community.
- Derivatives of benzene: Explain this term using an example from the book, your school, or your community.
- Polymers and polymerisation: Explain this term using an example from the book, your school, or your community.
- Solvent extraction and colligative properties: Explain this term using an example from the book, your school, or your community.
- Quantitative chemical equilibrium: Explain this term using an example from the book, your school, or your community.
- pH of acidic and alkaline solutions: Explain this term using an example from the book, your school, or your community.
- Indicators and titration curves: Explain this term using an example from the book, your school, or your community.
- Solubility and solubility product for sparingly soluble salts: Explain this term using an example from the book, your school, or your community.
- Electrochemical cell and applications: Explain this term using an example from the book, your school, or your community.
- Factors that affect the rate of reactions: Explain this term using an example from the book, your school, or your community.
- Rate laws and measurements: Explain this term using an example from the book, your school, or your community.
- Radioactivity: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Chemistry answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Chemistry that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.